Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad — July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering — *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi — Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering — *CGPA: 9.05*

Projects

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, FlashAttention-2, RoPE*

- Pretrained 25M-param decoder-only Small Language Models (SLMs) from scratch on 1B+ tokens across Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU.
- Trained dedicated 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility inflation and achieving 0.505 Bits-Per-Byte sequence compression across both high and low-resource languages.
- Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT via intermediate deductive scratchpads.

Harpy: Agentic AI Toolkit & Model Context Protocol Server | *Python, MCP, C++, CI/CD*

- Architected an open-standard JSON-RPC 2.0 stdio MCP server exposing 4 deterministic tools, connecting LLM coding agents (Cursor, Antigravity, Claude) to local C++ compiler toolchains and automated test runners.
- Engineered automated brute-force differential testing and a sandboxed multi-language runner with sub-millisecond execution benchmarking, TLE watchdog interrupts, and side-by-side terminal diffs.
- Implemented dual-channel synchronization via raw HTTP sockets and offline `.cph` test files, distributed as a PyPI package and PyInstaller standalone binary backed by Python 3.10–3.12 CI matrix testing.

AmbiStory: NLP Plausibility & Disambiguation Engine | *Python, PyTorch, Transformers, Embeddings*

- Achieved the best non-LLM solution on the AmbiStory problem (SemEval 2026), reaching $\rho = 0.535$ and 72.2% test accuracy.
- Engineered a 202-dim vector fusing multi-context sentence embeddings, NLI priors, and PCA interactions into an ordinal regression head.
- Boosted generalization by +0.265 ρ across unseen homonyms using group-stratified 5-fold CV, annotator KL divergence loss, and weight averaging to prevent out-of-distribution drift.

Football Video Analytics with YOLO | *Python, YOLOv8, ByteTrack, OpenCV, Streamlit, K-Means*

- Fine-tuned YOLOv8 with ByteTrack for multi-object tracking of players, referees, and the ball, leveraging deterministic K-Means clustering on jersey pixels for team classification and linear interpolation for ball-gap recovery.
- Compensated camera panning and perspective distortion via Lucas-Kanade optical flow and planar homography, mapping camera pixels to pitch coordinates to compute real-world sprint velocities, distance covered, and team ball possession.
- Re-architected pipeline to stream frames in batches with fused single-pass drawing and H.264 encoding, **cutting peak RAM from 5.3 GB to 1.0 GB**; built Streamlit app with content-hashed caching for instant re-runs and stats export.

Technical Skills

Languages: Python, SQL, C/C++, TypeScript, JavaScript, Bash.

Generative AI & Agentic Systems: LLMs & SLMs (Pretraining & SFT), Agentic AI, Model Context Protocol (MCP), FlashAttention-2, RoPE, SwiGLU, Chain-of-Thought, Tool Routing, Subprocess Sandboxing.

Machine Learning, NLP & CV: PyTorch, Hugging Face, Transformers (DeBERTa, MPNet), Scikit-learn, YOLOv8, OpenCV, K-Means Clustering, Optical Flow, SentencePiece, NLI.

MLOps, Data & Systems: Model Evaluation & Benchmarking, Docker, Linux/Unix, Git, CI/CD Pipelines, NumPy, Pandas, Matplotlib, Streamlit, Graph Algorithms (BFS, DFS).

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.